RETAIL SALES ANALYSIS

Final Project Report

Prepared by: Eshika Agarwal

Project: Internnova Data Analytics

1. Project Overview

This project analyzes the Sample Superstore retail dataset to understand sales, profit, quantity sold, category performance and regional trends. Python was used for data cleaning, exploratory data analysis (EDA) and visualizations. Power BI was used to build an interactive dashboard.

2. Dataset Description

The original dataset contained 9,994 records and 13 columns. After removing duplicate records, the cleaned dataset contained 9,977 records and 13 columns. The cleaned data was checked for missing values; the analysis reported no missing values after cleaning.

Item	Description
Dataset	Sample Superstore
Original records	9,994
Cleaned records	9,977
Columns	13
Key fields	Category, Sub-Category, Region, Sales, Profit, Quantity, Discount

3. Tools and Technologies

Python, Google Colab, Pandas, NumPy, Matplotlib, Power BI, Git and GitHub.

4. Methodology

1) Loaded the Sample Superstore CSV dataset in Google Colab.
2) Inspected the data structure and summary statistics.
3) Removed duplicate rows and standardized column names for analysis.
4) Checked missing values and performed exploratory analysis.
5) Created five visualizations in Python.
6) Imported the cleaned CSV into Power BI and created KPI cards, charts and slicers.
7) Summarized findings and prepared recommendations.

5. Visualizations and Dashboard

Python visualizations: Sales by Category; Profit by Category; Sales by Region; Quantity by Category; and Profit vs Discount. The Power BI dashboard contains KPI cards for Total Sales, Total Profit and Total Quantity, along with charts for category-wise sales, region-wise sales, profit by category and sales by sub-category. Region, Category and Segment slicers allow interactive filtering.

6. Key Business Insights

• Technology recorded the highest sales among the three categories (approximately 836,154.03).
• Technology also recorded the highest category profit (approximately 145,454.95).
• Furniture recorded the lowest category profit (approximately 18,421.81).
• The West region recorded the highest sales (approximately 725,255.64).
• The South region recorded the lowest sales (approximately 391,721.91).

7. Recommendations

1) Review pricing, discounting and operating costs in the Furniture category to improve its profitability.
2) Study customer preferences in the South region and use targeted marketing campaigns or suitable promotions toimprove sales.

8. Conclusion

The project demonstrates a basic end-to-end retail data analytics workflow, from dataset preparation and exploratory analysis to visualization and dashboard reporting. The findings highlight category and regional differences that can support business decision-making.

9. Project Files

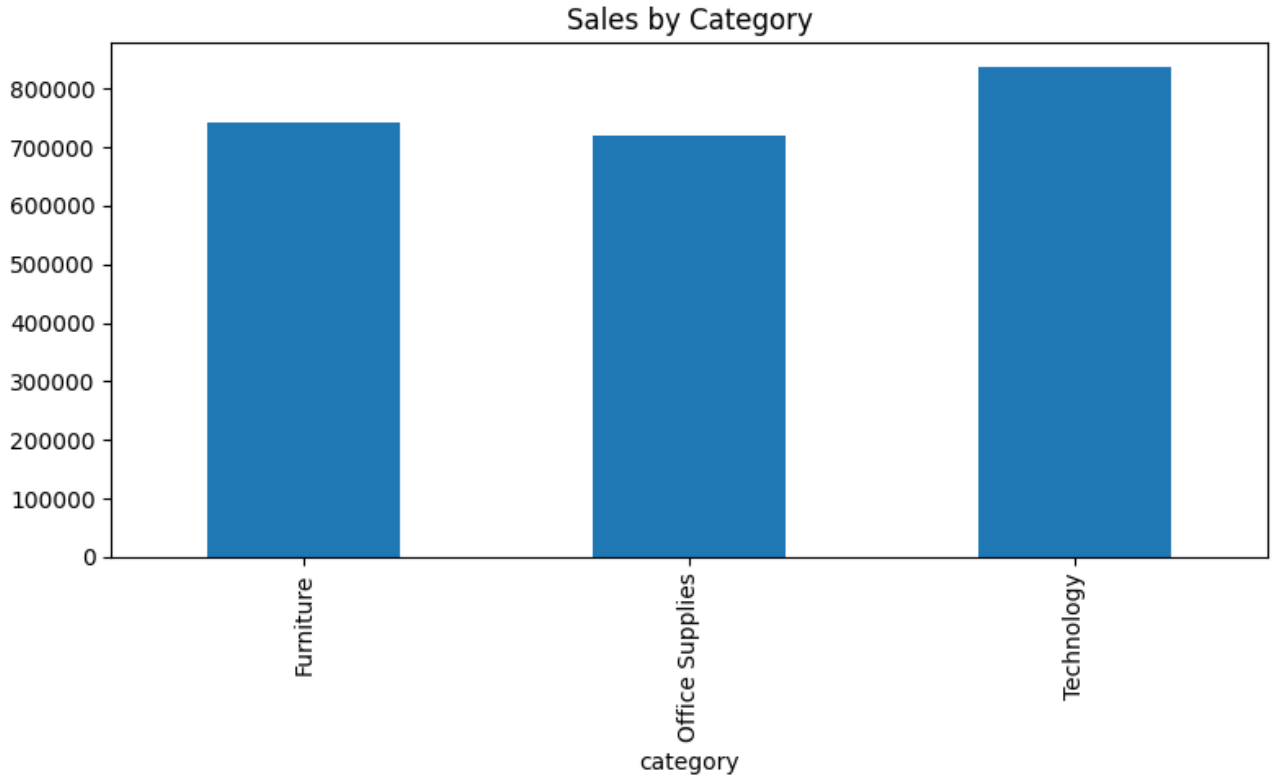
Data/retail_sales_cleaned.csv — cleaned dataset

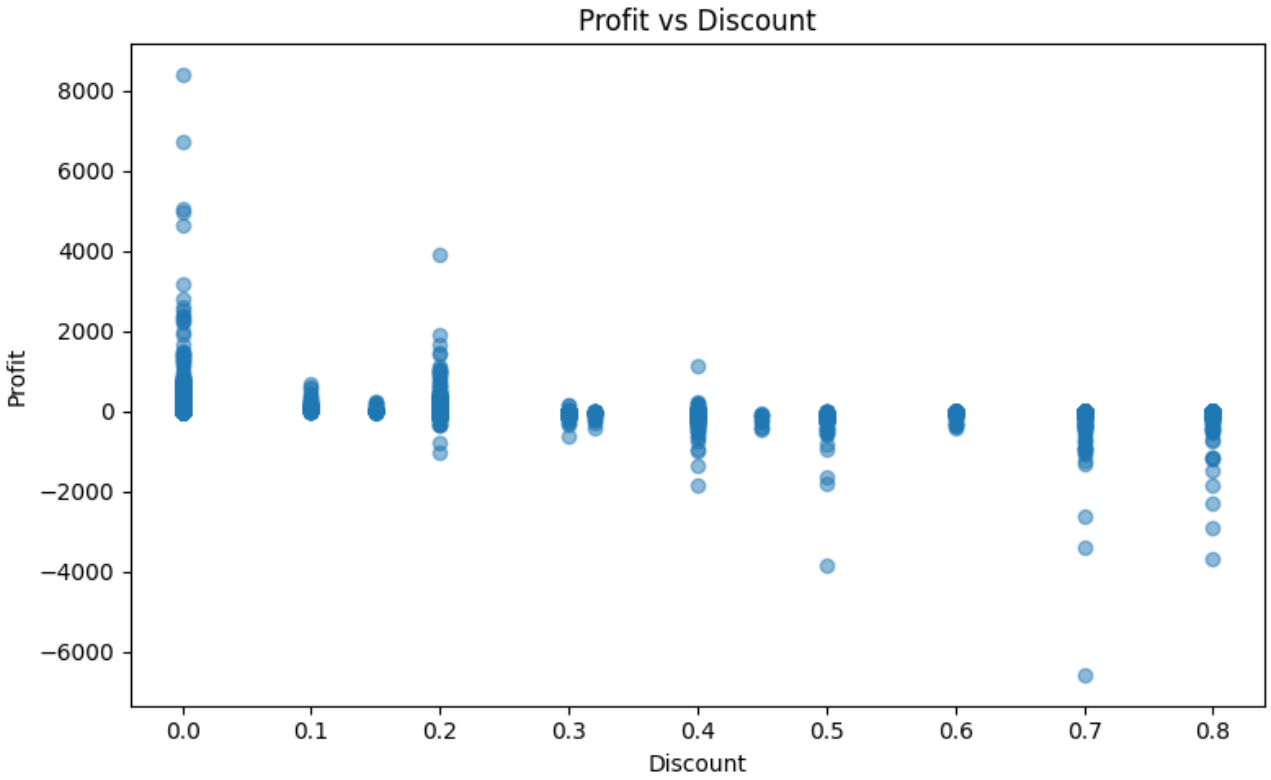
Notebooks/Retail_Sales_Analysis.ipynb — Python analysis notebook

PowerBi/Retail_Sales_Dashboard.pbix — Power BI dashboard

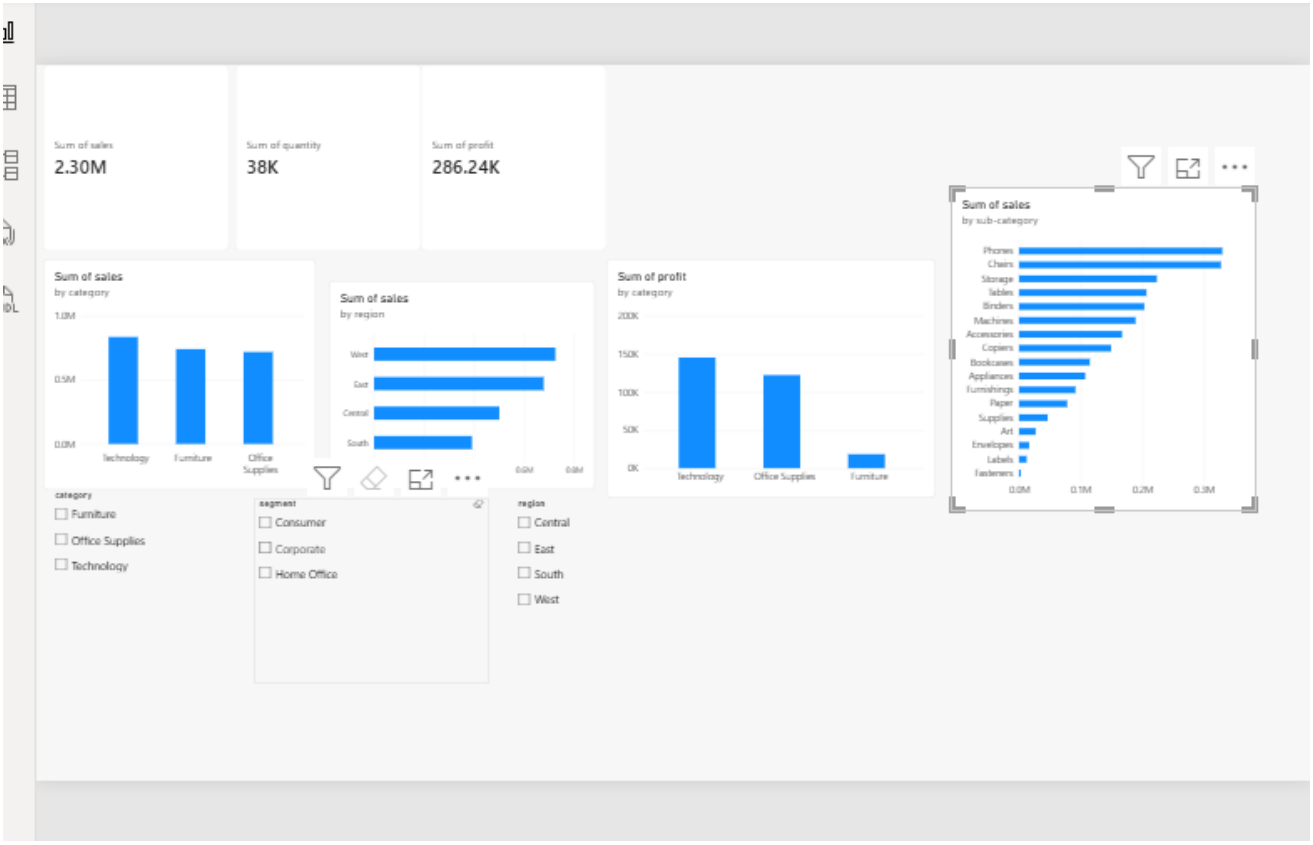
Visualization/ — five Python chart images
README.md — project documentation

Note: Values are based on the analysis outputs recorded during project development. Dashboard KPI totals may be displayed in rounded form.


Sales by Category
800000
700000
600000
500000
400000
300000
200000
100000
0
Furniture
Office Supplies
Technology
category



POWER BI DASHBOARD


Sum of sales
2.30M
Sum of quantity
38K
Sum of profit
286.24K
Sum of sales
by category
1.0M
0.5M
0.0M
Technology
Furniture
Office Supplies
Sum of sales
by region
West
East
Central
South
0.0M
0.1M
Sum of profit
by category
200K
150K
100K
50K
0K
Technology
Office Supplies
Furniture
Sum of sales
by sub-category
Phones
Chairs
Storage
Tables
Binders
Machines
Accessories
Copiers
Bookcases
Appliances
Furnishings
Paper
Supplies
Art
Envelopes
Labels
Fasteners
0.0M
0.1M
0.2M
0.3M
category
Furniture
Office Supplies
Technology
segment
Consumer
Corporate
Home Office
region
Central
East
South
West